School:_____ **Team number:**_____

Competitor Names: _____

Dynamic Planet

NORTH POCONO SCIENCE OLYMPIAD INVITATIONAL

Saturday, January 28, 2023

DO NOT OPEN PACKET UNTIL INSTRUCTED

Time Allowed:

50 min

Directions:

1. **Answer every question to the best of your ability. The written exam will be out of 100 points all multiple choice are 1 point and short answer points are listed next to each question.**
2. **Exams may be taken apart, but must be placed in CORRECT order when turned in.**
3. **Use correct significant figures where appropriate.**
4. **Use of a stand alone calculator of any type is permitted.**
5. **Answers MUST fit within the boxes provided. Additional scratch paper can also be provided during the exam. Any writing outside of the boxes provided, including that on scratch paper EVEN if it is referred to by writing inside of a box, e.g. "see last page"), will NOT be graded.**
6. **Calculation questions require showing of work for credit.**

Grading:

High score wins.

- Ties will be broken by:
    - The first team to answer a question incorrectly will lose the tie

Multiple Choice

1. What type of channel has an outer erosional surface and an inner depositional surface?
    a. Gauging
    b. Braided
    c. Meandering
    d. Straight

2. What weathering occurs when water seeps into the cracks of rocks and expands overtime to break the rock apart?
    a. Biological Weathering
    b. Exfoliation (Onion skin) Weathering
    c. Chemical Weathering
    d. Freeze-Thaw Weathering

3. _____ is the movement of water within spaces of porous material due to the forces of adhesion, cohesion, and surface tension.
    a. Capillary action
    b. Permeability
    c. Porosity
    d. Solution Valley

4. Where is the zone of saturation located?
    a. The grounds surface
    b. Immediately below the water table
    c. Immediately above the water table
    d. In bed rock

5. Wetlands formed from a river emptying it's water and sediment into another body of water.
    a. Deltas
    b. Waterfalls
    c. Dynamic Equilibrium
    d. Flood plains

6. In Image 2, identify the water table contour lines on the map
    a. Point C
    b. Point B
    c. Points A and C
    d. Points B and D

7. Which stream order method are ordered through additive means.
    a. Summation Method
    b. Computation Method
    c. Shreve Method
    d. Strahler Method

8. What is the best definition of a sink hole
    a. A dome-shaped mound into the ground consisting of a layer of soil over a large core.
    b. A cavity in the ground caused by water erosion and providing a route for surface drainage underground.
    c. A depression in the gound for water and stuff
    d. A cavity in the Earths surface that leads to a mysterious chamber full of secrets.

9. Identify: A type of pattern that develops from the disruption of a pre-existing patterns.
    a. Braided pattern
    b. Deranged pattern
    c. A drainage pattern
    d. Dendritic pattern

10. In hydrology and geomorphology what do they refer to “sea level” as?
    a. Equilibrium
    b. Sediment level
    c. Sediment source
    d. Base level

11. Which of the following is the best example of a nick point?
    a. Edge of a cliff
    b. Swamp
    c. Waterfall
    d. Highest point on mountain

12. These are all types of river/stream channels except _____
    a. Braided
    b. Centripetal
    c. Straight
    d. Meandering

13. Transport of small rocks and pebbles in rivers is called_____
    a. Saltation
    b. Sediment teleportation
    c. Traction
    d. Suspension

14. Which type of spring only flows at certain times of the year?
    a. Intermittent Springs
    b. Annual Springs
    c. Selective Springs
    d. Seasonal Springs

15. An area of flat land adjacent to a steam composed of sediment deposits by the steam is _____?
    a. Deposit Bank
    b. Delta
    c. Sediment Blank
    d. Flood Plain

16. Recurrence Intervals are usually measured in what?
    a. Years
    b. Centuries
    c. Miles
    d. Kilometers

17. Which of the following is not a way dams impact rivers and streams?
    a. Prevents fish migration
    b. Helps water quality
    c. Slows water flow
    d. Trap sediment where fish spawn

18. At what temperature and for how long does permafrost form?
    a. Continuous temp. of 0°F, for 2 or more years
    b. Fluctuating temp. of 20-40°C, for 3-9 months
    c. Continuous temp. of 0°C, for 2 or more years
    d. Fluctuating temp. of 20-40°F, for 3-9 months

19. What type of waterfall is in image 3A?
    a. Chute
    b. Block
    c. Cascade
    d. Plunge waterfall

20. What type of waterfall is in image 3B?
    a. Ribbon
    b. Plunge
    c. Block
    d. Punch bowl

21. What type of waterfall is in image 3C?
    a. Punch bowl
    b. Horse tail
    c. Tier
    d. Scree

22. What type of waterfall is in image 3D?
    a. Segmented
    b. Fan
    c. Tiered
    d. Horse tail

23. What is the difference between a levee and a dam?
    a. A levee is a wall of earth to control overflow of a river; a dam is a man made structure typically with water on one side and dry land on the other.
    b. A levee is a man made structure typically with water on one side and dry land on the other; a dam is a wall of earth to control overflow of a river.
    c. A levee is a tubular waterway made with natural materials; a dam is a barrier that stops or restricts the flow of surface water or underground streams.
    d. There is no difference

24. What is the sum of all processes that water moves from land to atmosphere?
    a. Evaporation
    b. The water cycle
    c. Precipitation
    d. Evapotranspiration

25. If a lakes shoreline is more jagged what is it more affected by?
    a. Runoff
    b. Inputs from surrounding land
    c. Groundwater leaching
    d. All of the above

26. Which are the best examples of inflow and outflow
    a. Inflow-Rainfall; Outflow-Precipitation
    b. Inflow-Runoff; Outflow-Rainfall
    c. Inflow-Rainfall; Outflow-Evaporation
    d. Inflow-Runoff; Outflow-Stratification

27. Where are seepage springs usually formed?
    a. At valley bottoms in sand and gravel
    b. At the bottom of depression in clay and bedrock
    c. At the top of hills in loose soil and rock
    d. On slopes after landslides

28. Which of the following in not a way surface water can be polluted?
    a. Oil spills from a ships
    b. Vehicle emissions
    c. Agriculture waste/runoff
    d. Sewage leaks

29. What is responsible for the influx of groundwater in aquifers?
    a. Discharge
    b. Capillary action
    c. Recharge
    d. Groundwater seepage

30. Which is the largest lake in the U.S.A. by area?
    a. Lake Ontario
    b. Lake Michigan
    c. Lake Superior
    d. Great Slat Lake

31. What type of cave is in image 4a?
    a. Eolian Cave
    b. Glacial Cave
    c. Primary Cave
    d. Talus Cave

32. What type of cave is in image 4b?
    a. Eolian Cave
    b. Solution Cave
    c. Talus Cave
    d. Lava Cave

33. What type of cave is in image 4c?
    a. Coastal Cave
    b. Sea Cave
    c. Eolian Cave
    d. Solution Cave

34. What type of cave is in image 4d?
    a. Sea Cave
    b. Coastal Cave
    c. Rock shelter
    d. Solution cave

35. What is the zone of aeration?
    a. The region between Earths surface and the water table made up of mostly soil and rocks
    b. The region between the water table and bedrock made up of rocks and gravel
    c. Where saltwater and freshwater meet in groundwater
    d. The region immediately below the water table

36. The formation of the Sea of Galilee, in the Jordan valley, is an example of what type of lake?
    a. Rift Lake
    b. Lava-Blocked Lake
    c. Kettle Lake
    d. Crater Lake

37. Which of the following is not a type of glacial lake?
    a. Kettle Lakes
    b. Cirque Lakes
    c. Rock-Hollow Lakes
    d. Caldera Lakes

38. What are flow lines?
    a. Represent the flow of surface water through valleys
    b. The lines on a map that show slopes
    c. Represent the path which water will seep through soil
    d. The lines that represent the valleys on a map

39. What is the largest freshwater lake by volume?
    a. Lake Michigan
    b. Lake Baikal
    c. Caspian Sea
    d. Lake Tanganyika

40. This drainage pattern makes sharp bends and enter the main stream at high angles.
    a. Radical Drainage
    b. Rectangular Drainage
    c. Deranged Drainage
    d. Trellis Drainage

41. This drainage pattern is characteristic of folded mountains.
    a. Trellis Drainage
    b. Radical Drainage
    c. Parallel Drainage
    d. Dendritic Drainage

42. This drainage originates from a hill with flow in all directions
    a. Centripetal Drainage
    b. Trellis Drainage
    c. Parallel Drainage
    d. Radial Drainage

43. If a stream's load is mostly gravel and sand, what type of channel would it have?
    a. Meandering
    b. Straight
    c. Braided
    d. Twisted

44. In saltwater intrusion, does the saltwater primarily penetrate?
    a. Horizontally
    b. Vertically
    c. Parallel
    d. Laterally

45. According to Darcy's law, when does ground water flow faster?
    a. When infraction increases and runoff decreases
    b. When the slope of the water table creates high hydraulic head
    c. When the water temperature changes drastically
    d. When the sediment erodes completely

46. What is the hydraulic conductivity (k) in ft/day is sand?
    a. 10^{-4} - 10^{-6}
    b. 10 - 10^{2}
    c. 1 - 10
    d. 10^{-6} - 10^{-4}

47. These are found along shallow edges of lakes and rivers.
    a. Swamps
    b. Bogs
    c. Wet prairie
    d. Marshes

48. Where are bogs usually found?
    a. In cool northern climates
    b. In the mountains towards the west
    c. Near swamps in the south
    d. In the hot sandy deserts

49. What is lake stratification?
    a. A form of transport for sediment in rivers
    b. The limit below which a stream cannot erode
    c. The tendency of lakes to form distinct thermal layers during warm weather
    d. To make the balance in water flow and sediment transport

50. When wind blows over a calm lake what are the first type of waves to form?
    a. Intermediate Waves
    b. Surface Waves
    c. White-cap Waves
    d. Wind-surge Waves

51. What is the salinity of a lake?
    a. How much saline solution is in a lake
    b. A test run to see if the water is sanitary
    c. To tell if the lake is a saline lake
    d. How much salt is in it

Short Answer

52. In Image 1 when traveling from point A to point B assuming you move in a straight line. How will the elevation change as you travel? (5 points)
    As you travel your elevation will first increase when hicking up the Mount. Marcy, then decrease going down the mountain and lastly increase by climbing up Panther Gorge peak.

53. Explain what type of structure are the Pueblo Indian cliff dwellings in Colorado(Mesa Verde)? (5 points)
    The Mesa Verde cliff dwellings are an example of rock shelters, a type of cave. Rock shelters are produced when bedrock erosion takes place in soluble rocks like sandstone. In our case the Pueblo Indians found rock structures on the sides of cliffs and established their homes there while still farming on the Mesa tops.

54. What type of waterfall is Niagara falls? (2 points)
    Niagara falls is a block water fall **OR** it can also be identified as a horseshoe waterfall

55. What are the 5 main processes that make up the hydrologic cycle? (2 points)
    Evaporation, transpiration, condensation, precipitation, and runoff.

56. Label the boxes in image 5. (4 points)
    A. Disappearing Stream
    B. Sinkholes
    C. River
    D. Reappearing streams (will accept streams)
    E. Groundwater Table
    F. Solution Valley
    G. Collapsed sinkhole
    H. Cave

57. Explain how glacial lakes are formed. (2 points)
    Most glacial lakes form when a glacier retreats and meltwater fills the hole left behind.

58. Labe 1-5 on image 6 (5 points)
    1. Surface water
    2. Underground water
    3.Unsaturated zone
    4. Saturated zone
    5. Intermediate zone

59. Label 6-10 on image 6 (5 points)
    6. Soil zone
    7. Water level
    8. Ground water
    9. Water table
    10. Capillary Fringe

60. Identify the classes of sinuosity of the following:(3 points)
    SI <1.05 :
    1.05 ≤ SI <1.25 :
    1.25 ≤ SI <1.50 :
    1.50 ≤ SI :
    SI <1.05: almost straight, 1.05 ≤ SI <1.25: winding, 1.25 ≤ SI <1.50: twisty, 1.50 ≤ SI: meandering

61. How can wetlands improve water quality? (3 points)
    They can improve water quality by removing pollutants from surface waters. Three pollutant removal processes provided by wetlands are particularly important: sediment trapping, nutrient removal and chemical detoxification.

62. Label the thermal stratification layers in Image 7. (3 points)
    A. Epilimnion
    B. Metalimnion
    C. Hypolimnion

63. What is the purpose/function of vegetation in wetlands? (4 points)
    Wetland vegetation provides food and critical habitat for organisms that live in or near water resources, such as algae, macroinvertebrates, amphibians, fish and birds. Wetland plants can also improve water quality through the uptake of nutrients, metals and other contaminants.

64. What type of unique draining is found primarily in the Appalachian Mountains? (2 points)
    Trellis Drainage

65. Lakes undergo changes over time. Just as humans age, lakes can age. What are some of the effects of aging lakes? (4 points)
    Lakes slowly disappear as their basins fill with sediment and plant material as plants and algae slowly die, this happens over 100's and 1,000's of years. This is sped up to only take decades through human influence.